1). Write a HTML/5 program to demonstrate the use of Font family, font variant, font style, and font size.

```
<!DOCTYPE html>
<html>
<head>
<title>font properties, like  size, style, colors</title>
</head>
<body>
<h1>Font Size 1</h1>
<h2>Heading 2</h2>
<h3>Heading 3</h3>
<h1>Font style 1</h1>
<p>This text is normal.</p>
<p><i>This text is italic.</i></p>
<h1>Font Colors 1</h1>
<h1 style="color:blue;">Guru Nanak first Grade College Bidar</h1>
<p style="color:red;">BCA 4th Sem Students</p>
</body>
</html>
```

Output:

Font Size 1

Heading 2

Heading 3

Font style 1

This text is normal.

This text is italic.

Font Colors 1

Guru Nanak first Grade College Bidar

BCA 4th Sem Students

2)Write a HTML/5 program to display random contents using list properties:
a) Ordered list b) Unordered list

```
<!DOCTYPE html>
<html>
 <head>
 <title>HTML Ordered and un-ordered List</title>
   </head>
   <body>
   <h2>List of Fruits</h2>
     <ol>
       <li>Apple</li>
       <li>Mango</li>
       <li>Banana</li>
       <li>Grapes</li>
       <li>Orange</li>
     </ol>
   <h2>List of Fruits</h2>
    <ol type="A">
       <li>Apple</li>
       <li>Mango</li>
      <li>Banana</li>
   </ol>
   </body>
</html>
```

Output:

List of Fruits

1. Apple
2. Mango
3. Banana
4. Grapes
5. Orange

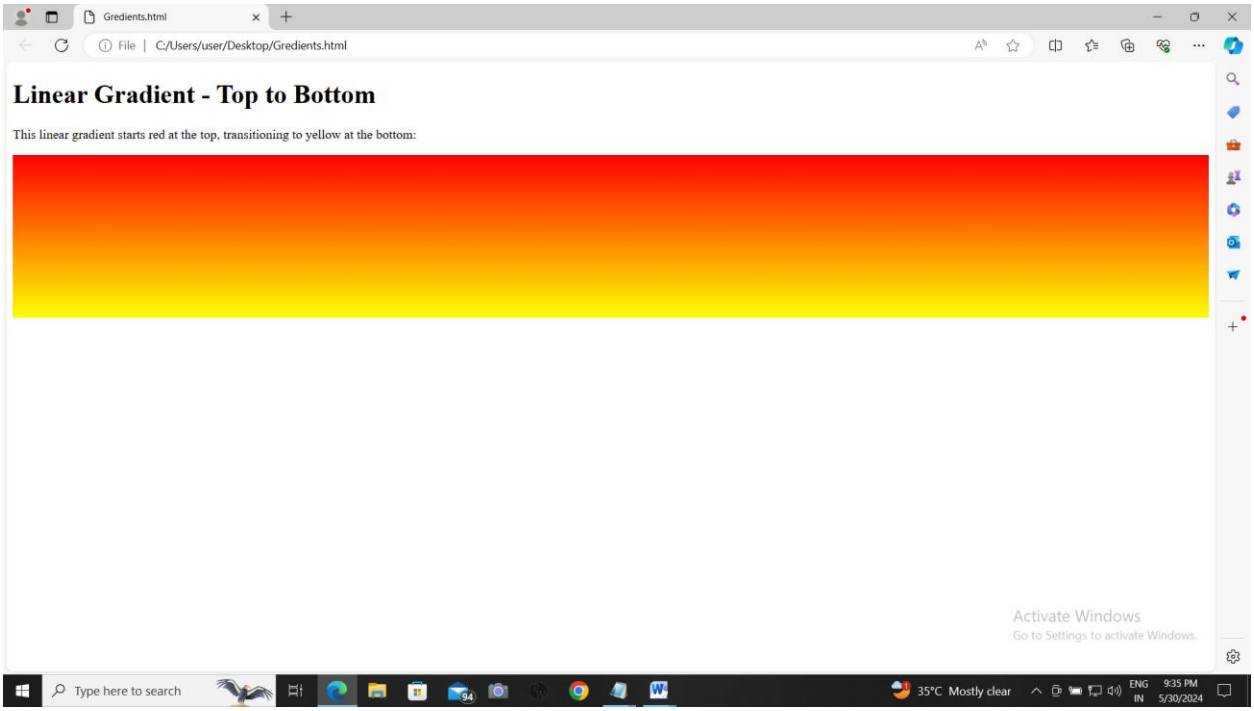
List of Fruits

A. Apple
B. Mango
C. Banana

3)Write a HTML/5 program to create gradient using CSS.

```
<!DOCTYPE html>
<html>
<head>
<style>
#grad1
 {
  height: 200px;
  background-color: red; /* For browsers that do not support gradients */
  background-image: linear-gradient(red, yellow);
}
</style>
</head>
<body>
<h1>Linear Gradient - Top to Bottom</h1>
<p>This linear gradient starts red at the top, transitioning to yellow at the bottom:</p>
<div id="grad1"></div>
</body>
</html>
```

Output:

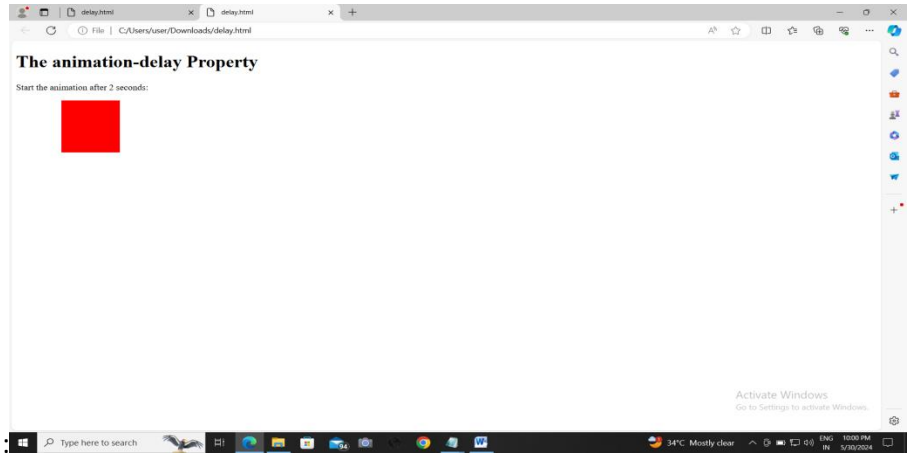


4) Write a HTML/5 code to demonstrate following CSS animation properties: a) Delay b) Direction c) Duration

a) Delay

```
<!DOCTYPE html>
<html>
<head>
<style>
div {
  width: 100px;
  height: 100px;
  background: red;
  position: relative;
  animation: mymove 5s infinite;
  animation-delay: 2s;
}
@keyframes mymove {
  from {left: 0px;}
  to {left: 200px;}
}
</style>
</head>
<body>
<h1>The animation-delay Property</h1>
<p>Start the animation after 2 seconds:</p>
<div></div>
</body>
</html>
```

Output:

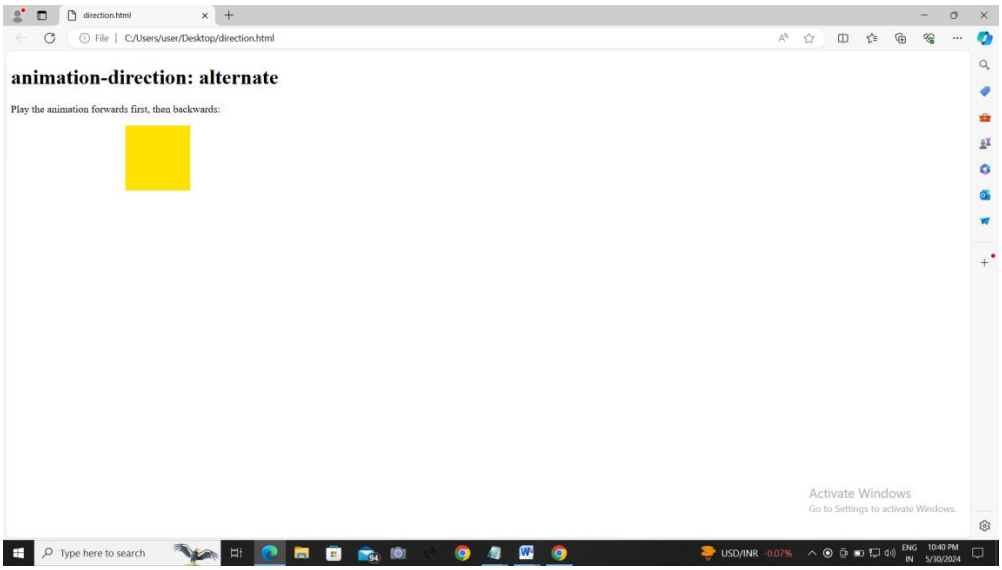


4) Write a HTML/5 code to demonstrate following CSS animation properties:

b) Direction

```
<!DOCTYPE html>
<html>
<head>
<style>
div {
  width: 100px;
  height: 100px;
  background: red;
  position: relative;
  animation: myfirst 3s 2;
  animation-direction: alternate;
}
@keyframes myfirst {
  0%   {background: red; left: 0px; top: 0px;}
  25%  {background: yellow; left: 200px; top: 0px;}
  50%  {background: blue; left: 200px; top: 200px;}
  75%  {background: green; left: 0px; top: 200px;}
  100% {background: red; left: 0px; top: 0px;}
}
</style>
</head>
<body>
<h1>animation-direction: alternate</h1>
<p>Play the animation forwards first, then backwards:</p>
<div></div>
</body>
</html>
```

Output:

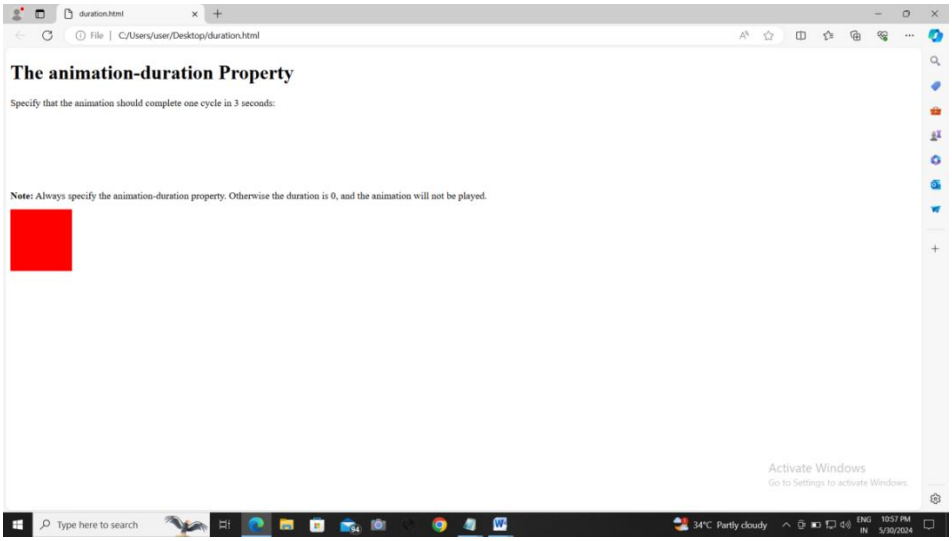


4) Write a HTML/5 code to demonstrate following CSS animation properties:

c) Duration

```
<!DOCTYPE html>
<html>
<head>
<style>
div {
  width: 100px;
  height: 100px;
  background: red;
  position: relative;
  animation: mymove infinite;
  animation-duration: 3s;
}

@keyframes mymove {
  from {top: 0px;}
  to {top: 200px;}
}
</style>
</head>
<body>
<h1>The animation-duration Property</h1>
<p>Specify that the animation should complete one cycle in 3 seconds:</p>
<div></div>
<p><strong>Note:</strong> Always specify the animation-duration property. Otherwise the duration is 0, and the animation will not be played.</p>
</body>
</html>
```

Output:



5)Write a HTML/5 program to demonstrate key frames

```
<!DOCTYPE html>
<html>
<head><Title>key frame</Title>
<style>
h1 {
  letter-spacing: 3px;
}
h2 {
  letter-spacing: 2px;
}
h3 {
  letter-spacing: -1px;
}
</style>
</head>
<body>
<h1>This is heading 1</h1>
<h2>This is heading 2</h2>
<h3>This is heading 3</h3>
</body>
</html>
```

Output:

This is heading 1

This is heading 2

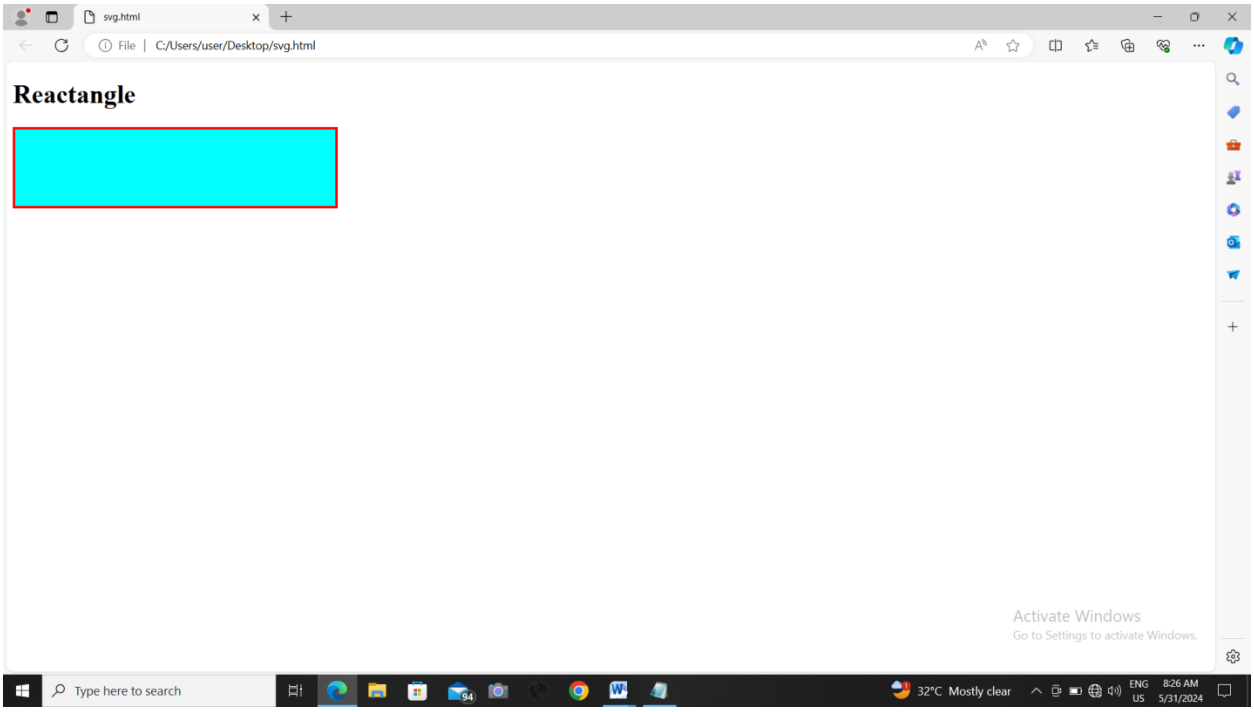
This is heading 3

6)Write a HTML/5 program to draw rectangle, line, polygon, polyline using SVG.

a) Write a HTML/5 program to draw rectangle using SVG.

```
<!DOCTYPE html>
<html>
<body>
<h1>Reactangle</h1>
  <svg width="400" height="100">
    <rect width="400" height="100"
    style="fill:rgb(0, 255, 255);
    stroke-width:6;stroke:rgb(255, 0, 0)" />
  </svg>
</body>
</html>
```

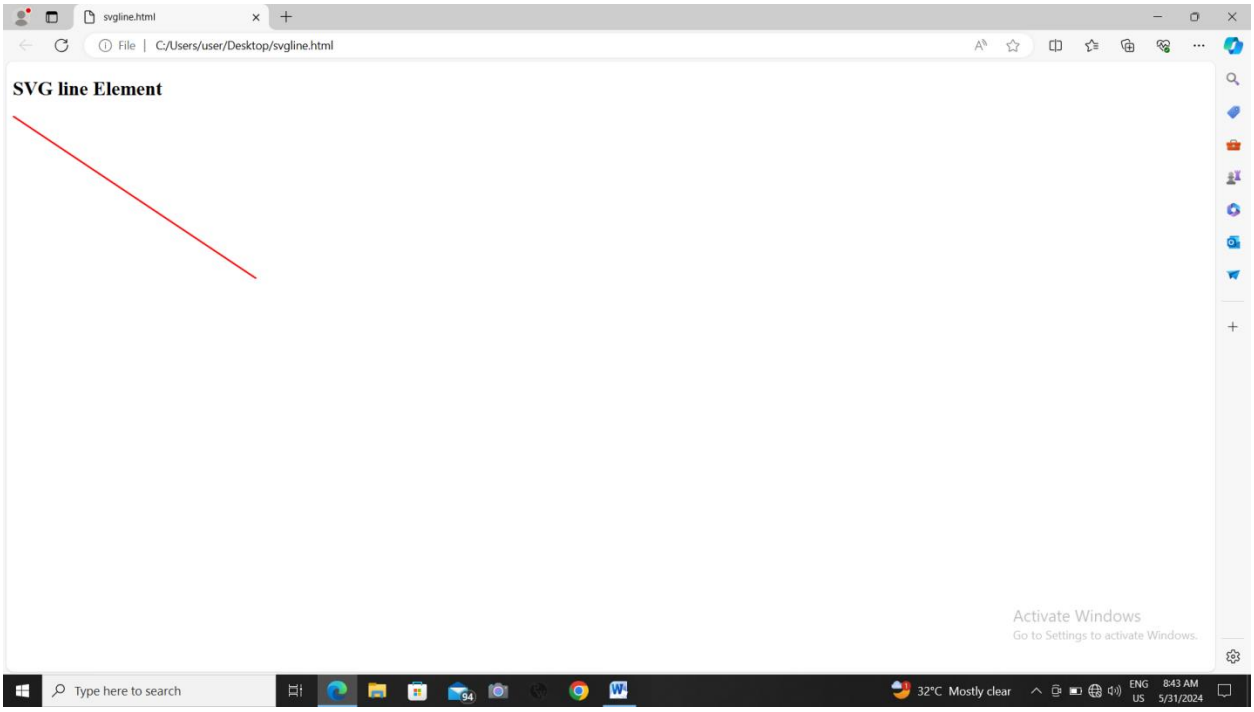
Output:



b)**Write a HTML/5 program to draw line using SVG.**

```
<!DOCTYPE html>
<html>
<body>

<h2>SVG line Element</h2>

<svg height="200" width="300" xmlns="http://www.w3.org/2000/svg">
  <line x1="0" y1="0" x2="300" y2="200" style="stroke:red;stroke-width:2" />
  Sorry, your browser does not support inline SVG.
</svg>

</body>
</html>
```

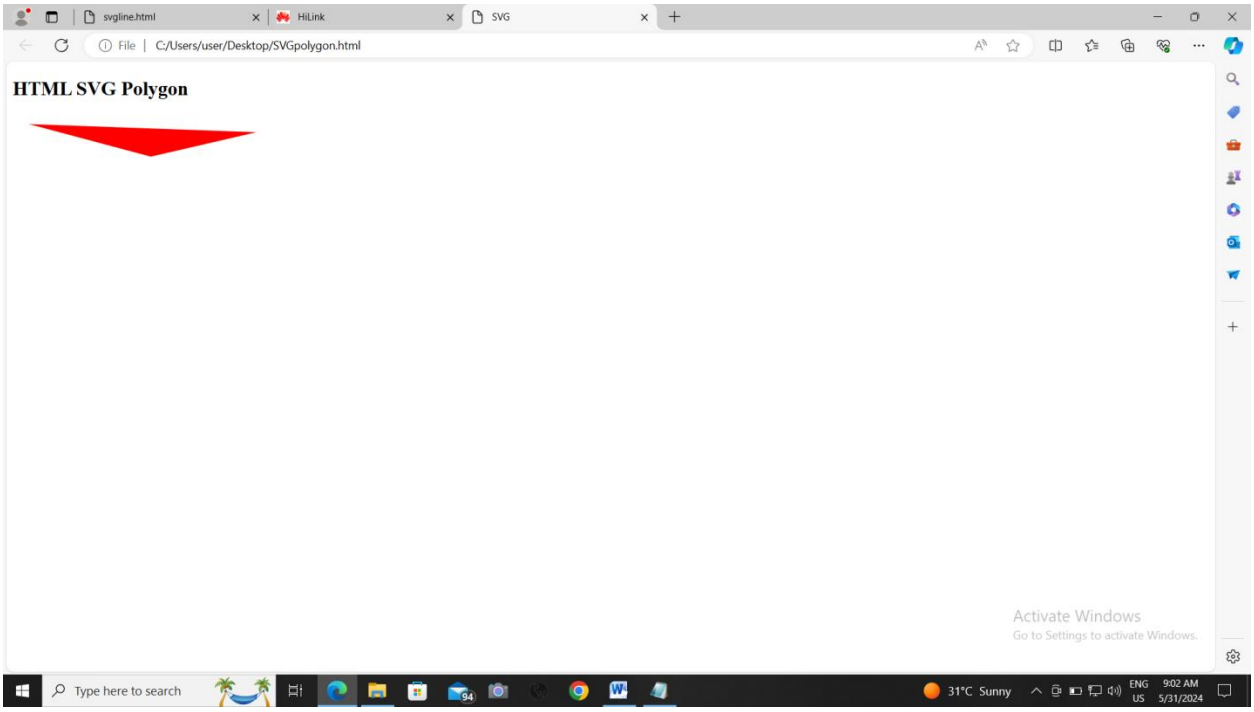
Output:



c).**Write a HTML/5 program to draw polygon using SVG.**

```
<!DOCTYPE html>
<html>
<head>
  <title>SVG</title>
  <meta charset="utf-8" />
</head>
<body>
  <h2>HTML SVG Polygon</h2>
  <svg id="svgelem" height="200" xmlns="http://www.w3.org/2000/svg">
    <polygon points="20,10 300,20, 170,50" fill="red" />
  </svg>
</body>
</html>
```

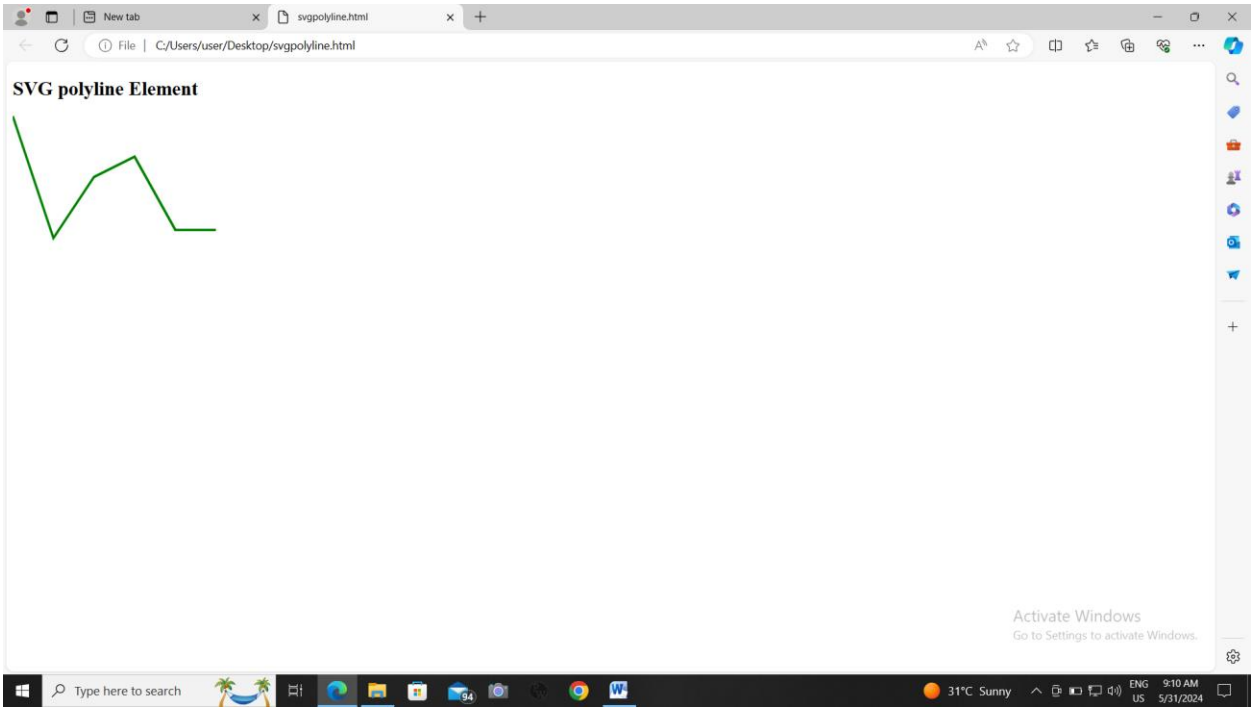
Output:



d**).Write a HTML/5 program to draw polyline using SVG.**

```
<!DOCTYPE html>
<html>
<body>

<h2>SVG polyline Element</h2>

<svg height="210" width="500" xmlns="http://www.w3.org/2000/svg">
  <polyline points="0,0 50,150 100,75 150,50 200,140 250,140" style="fill:none;stroke:green;stroke-width:3" />
</svg>

</body>
</html>
```

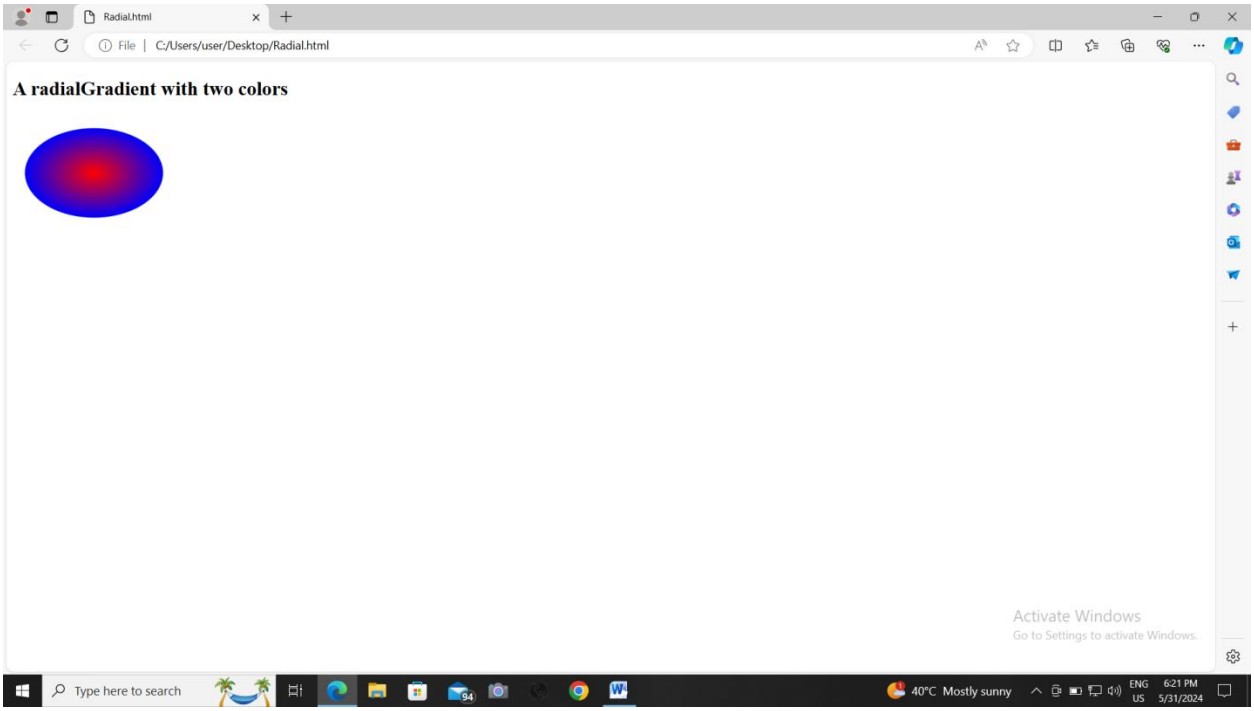
Output:



7).Write a HTML/5 program to draw Radial gradient ellipse using SVG.

```
<!DOCTYPE html>
<html>
<body>

<h2>A radialGradient with two colors</h2>

<svg height="150" width="400" xmlns="http://www.w3.org/2000/svg">
  <defs>
    <radialGradient id="grad1" cx="50%" cy="50%" r="50%" fx="50%" fy="50%">
      <stop offset="0%" stop-color="red" />
      <stop offset="100%" stop-color="blue" />
    </radialGradient>
  </defs>
  <ellipse cx="100" cy="70" rx="85" ry="55" fill="url(#grad1)" />

</svg>

</body>
</html>
```

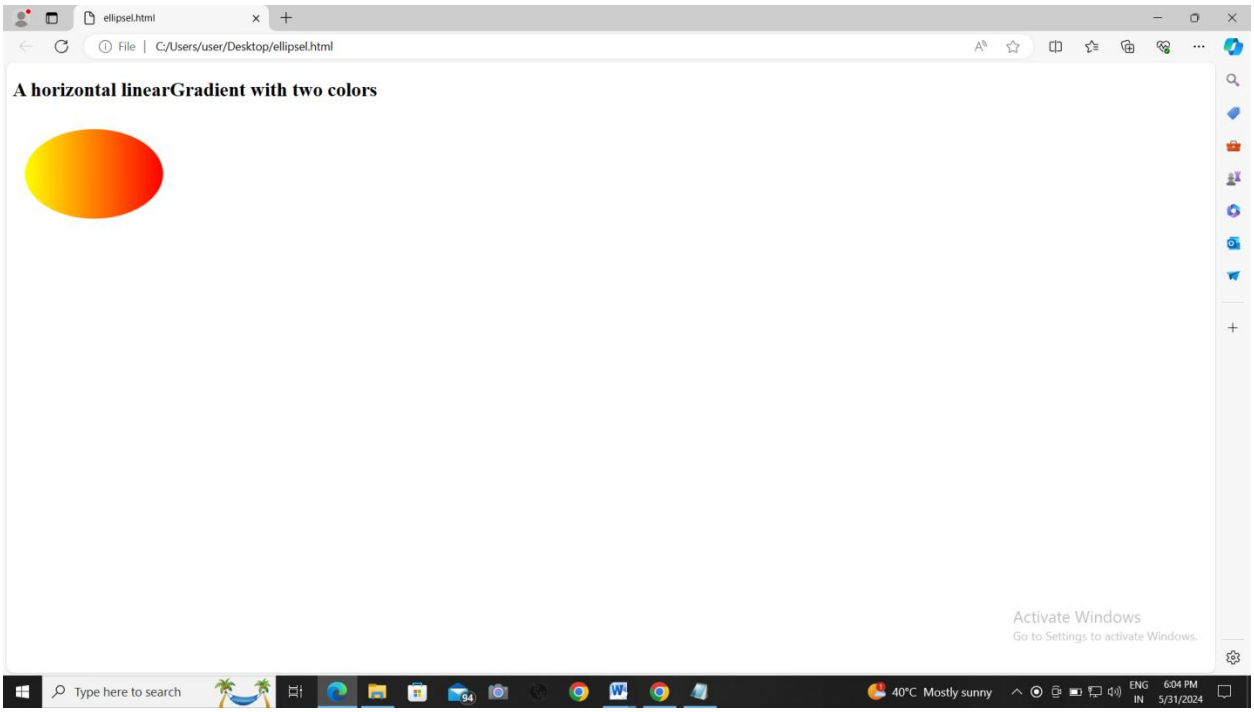
Output:



8).Write a HTML/5 program to draw linear gradient ellipse using SVG.

```
<!DOCTYPE html>
<html>
<body>

<h2>A horizontal linearGradient with two colors</h2>

<svg height="150" width="400" xmlns="http://www.w3.org/2000/svg">
  <defs>
    <linearGradient id="grad1" x1="0%" x2="100%" y1="0%" y2="0%">
      <stop offset="0%" stop-color="yellow" />
      <stop offset="100%" stop-color="red" />
    </linearGradient>
  </defs>
  <ellipse cx="100" cy="70" rx="85" ry="55" fill="url(#grad1)" />

</svg>

</body>
</html>
```

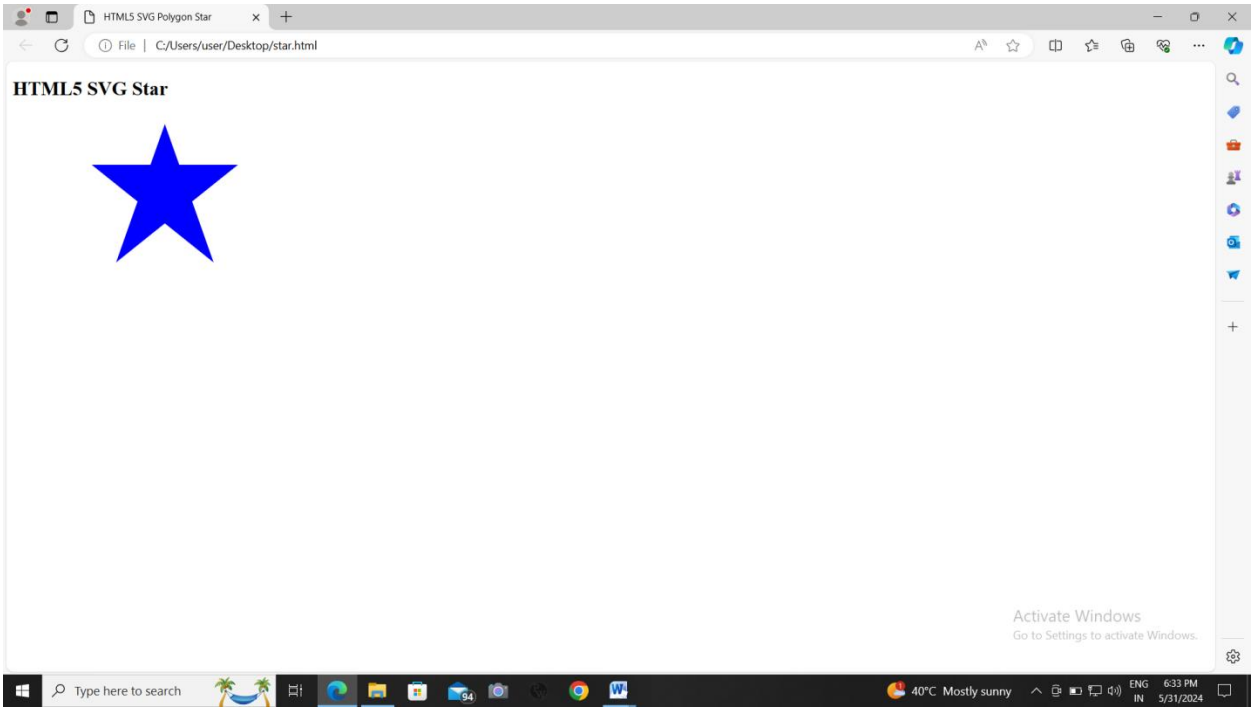
Output:



9)Write a HTML/5 program to draw a star using SVG.

```
<!DOCTYPE html>
  <html>
    <head>
      <style>
        #svgelem {
          position: relative;
          left: 10%;
          -webkit-transform: translateX(-20%);
          -ms-transform: translateX(-20%);
          transform: translateX(-20%);
        }
      </style>
    <title>HTML5 SVG Polygon Star</title>
  </head>
  <body>
    <h2>HTML5 SVG Star</h2>
      <svg id = "svgelem" width = "300" height = "300" xmlns = "http://www.w3.org/2000/svg">
        <polygon points="100,10 40,180 190,60 10,60 160,180" fill="blue"/>
      </svg>
  </body>
</html>
```

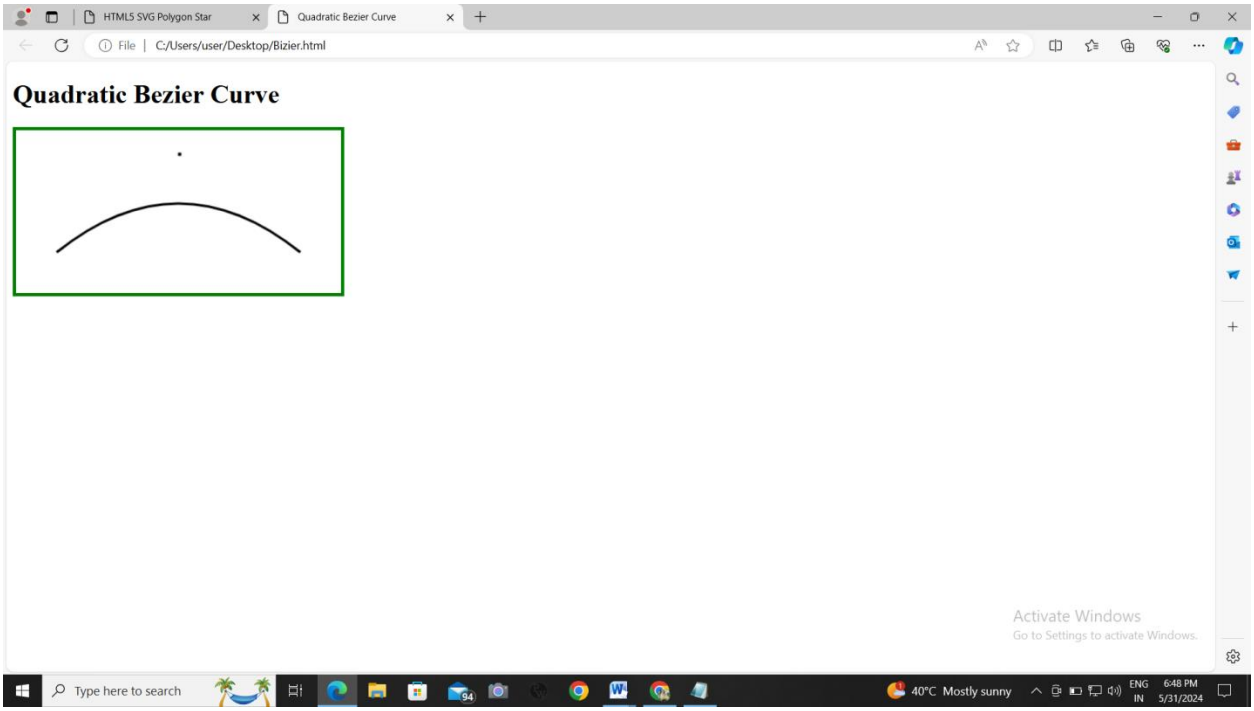
Output:



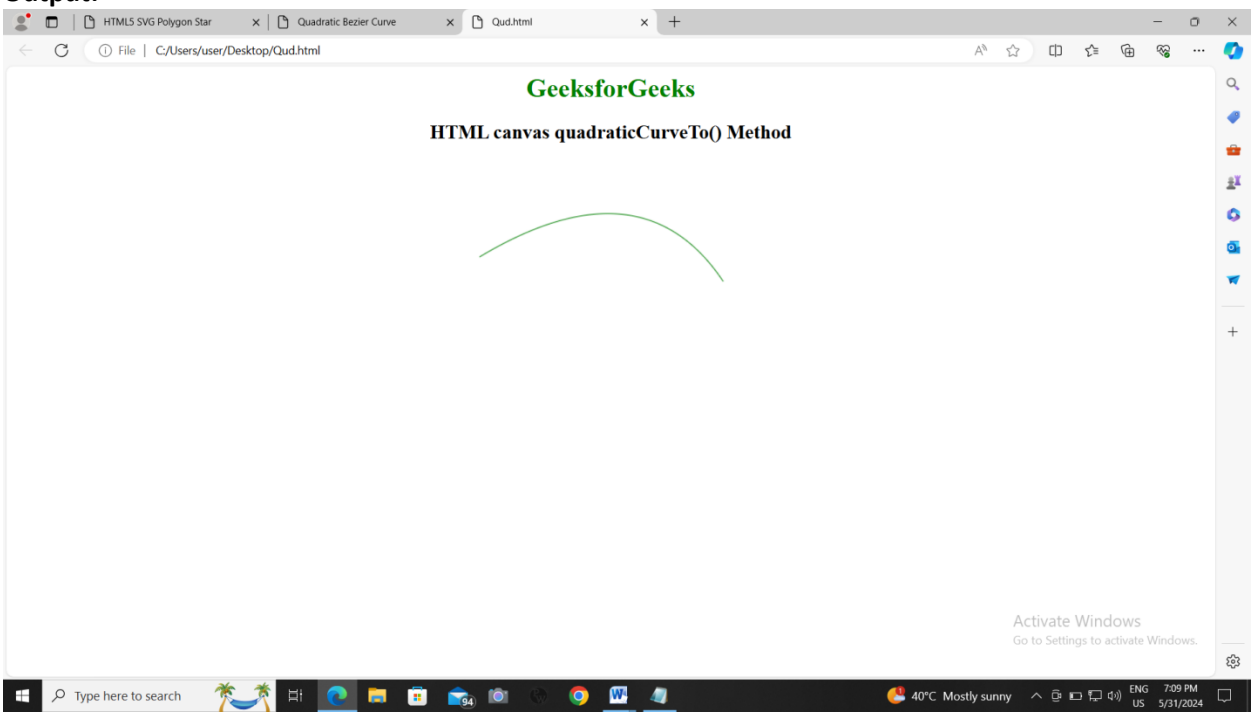
10).Write a HTML/5 program to demonstrate Bezier Curves and Quadratic Curves.

```
<!DOCTYPE html>
<html>

<head>
   <title>
      Quadratic Bezier Curve
   </title>
</head>

<body>
   <h1>Quadratic Bezier Curve</h1>

   <canvas id="CanvasOfGeeks" width="400" height="200"
         style="border:solid 4px green">

   <script>
      var c = document.getElementById("CanvasOfGeeks");
      var context = c.getContext("2d");

      var start_pnt_X = 50;
      var start_pnt_Y = 150;
      var cntrl_pnt_X = 200;
      var cntrl_pnt_Y = 30;
      var end_pnt_X = 350;
      var end_pnt_Y = 150;

      /* Start new path */
      context.beginPath();
      context.lineWidth=3;
      context.strokeText( ".", cntrl_pnt_X, cntrl_pnt_Y);

      /* Starting point of the curve */
      context.moveTo(start_pnt_X, start_pnt_Y);

      context.quadraticCurveTo(cntrl_pnt_X,
               cntrl_pnt_Y, end_pnt_X, end_pnt_Y);

      /* drawing line on the canvas */
      context.stroke();
   </script>
</body>

</html>
```

Output:


HTML5 SVG Polygon Star
Quadratic Bezier Curve
File | C:/Users/user/Desktop/Bizier.html
Quadratic Bezier Curve
Activate Windows
Go to Settings to activate Windows.
Type here to search
40°C Mostly sunny
ENG IN
6:48 PM 5/31/2024

Write a HTML/5 program to demonstrate Quadratic Curves.

```
<body style="text-align:center;">

  <h1 style="color:green;">
    GeeksforGeeks
  </h1>

  <h2>HTML canvas quadraticCurveTo() Method</h2>

  <canvas id="GFG" width="500" height="300"></canvas>

  <script>
    var geeks = document.getElementById('GFG');
    var context = geeks.getContext('2d');
    context.beginPath();
    context.moveTo(88, 120);
    context.quadraticCurveTo(288, 0, 388, 150);
    context.strokeStyle = 'green';
    context.stroke();
  </script>
</body>

</html>
```

Output: